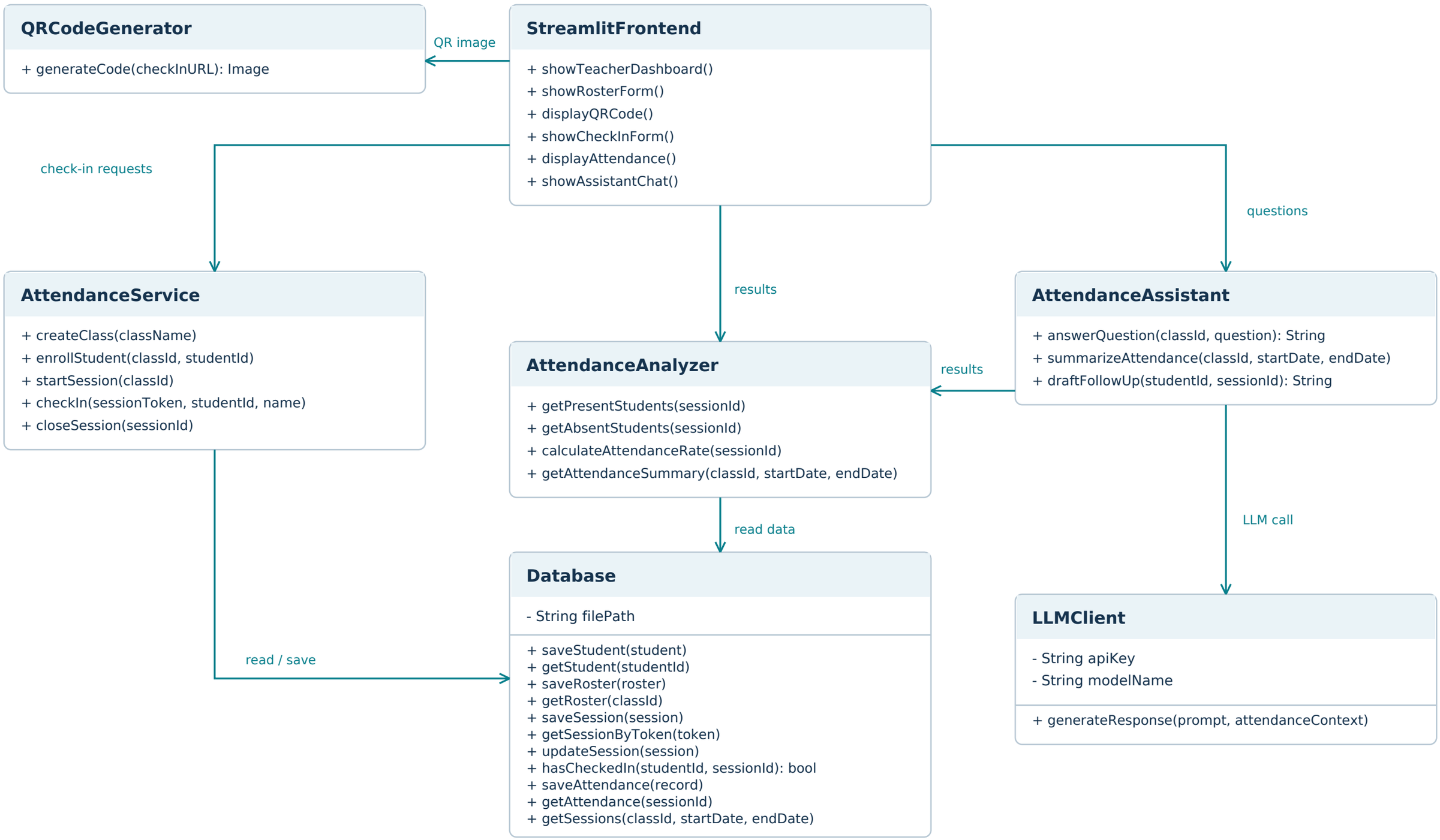
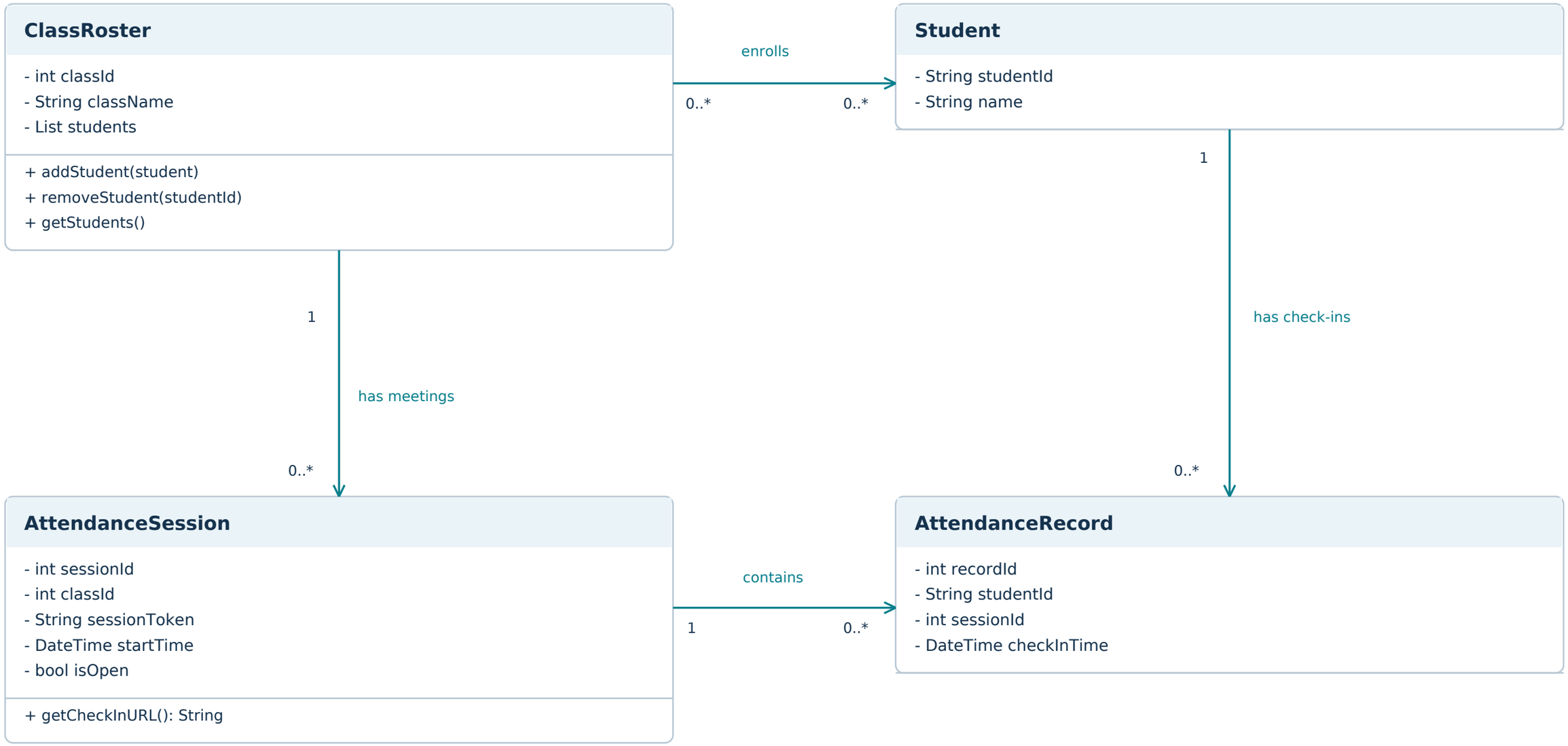
Application and service UML

The front end delegates check-in, analysis, QR generation, and natural-language questions to separate components.



Domain class UML

A session belongs to one class. Each successful check-in links one enrolled student to one session.



Service dependencies

AttendanceService manages ClassRoster, creates and closes AttendanceSession, and creates AttendanceRecord.

Legend: + public method | - private field | 1 one | 0..* zero or many | arrows show navigation or use.

Responsibilities and operating rules

Implementation notes for the complete QR attendance system.

StreamlitFrontend	Teacher dashboard, roster entry, QR display, student check-in form, and assistant chat.
AttendanceService	Creates classes and sessions; verifies enrollment, open sessions, and duplicate check-ins.
Student / ClassRoster	Stores student identity and the enrolled students for each class.
AttendanceSession	Represents one class meeting and its unique check-in link.
QRCodeGenerator	Converts the session URL into a QR image the teacher can display or share.
AttendanceRecord	Stores one successful check-in, including the student, session, and time.
AttendanceAnalyzer	Compares check-ins with the roster and calculates attendance statistics.
AttendanceAssistant	Gets calculated attendance results and asks the LLM to explain or summarize them.
LLMClient	Calls Gemini or another LLM. Keeps its API key on the server.
Database	Stores students, class enrollments, sessions, and attendance records in attendance.db.

Check-in and analysis rules

1. Each QR code points to the check-in page for one attendance session.
2. Accept check-in only while the session is open and the student is enrolled in that class.
3. Enforce UNIQUE(studentId, sessionId) in SQLite to prevent duplicate check-ins, including simultaneous submissions.
4. While a session is open, missing students are "not checked in yet". After it closes, they are absent.
5. Python retrieves records, matches student IDs, and calculates totals. The LLM explains those results and drafts messages.
6. A typed name and ID do not verify identity. School login is a possible future addition for verified check-in.

Python organization

These components can be Python classes or modules. The Streamlit front end can be app.py with functions.